A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 30

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Hydrosere **1** Single-celled algae (and floating plants) - arrive on birds / insects / from soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pioneer species** → the first species to colonise bare ground	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 5 points below. **1** Pioneers and later plants grow and die – dead organic matter plus sediment washed in fill the pond so water becomes shallower **1** As depth falls rooted plants (pondweeds then emergent reeds with leaves above water) can anchor and colonise – emergents shade submerged plants which die out **1** Sediment / soil accumulates and becomes able to support larger plants – water-tolerant trees (willow, alder) colonise the waterlogged soil **1** Transpiration by trees and further sediment dry the soil and make it aerobic so oak can colonise – the denser canopy shades out smaller plants **1** Each community changes abiotic conditions (depth, light, nutrients, oxygen), making it suitable for the next colonisers which out-compete them – conditions become less extreme and biotic factors increasingly important **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogged** → soil pores filled with water so there is no air • **Transpiration** → water lost as vapour from the leaves • **abiotic** → non-living / physical factors	5
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Starting conditions (bare rock vs open water) differ but each stage makes conditions less extreme and builds soil / organic matter **1** The final community is controlled by the regional climate (temperature, water availability), which is the same for both - climatic climax (eg oak woodland in the UK) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **climax community** → the stable final community **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	2
01.4	1 mark for the point below. **1** Lichens and algae colonise bare rock (extreme temperatures, no soil, little water) – rock fragments and dead organic matter accumulate – mosses, then grasses and ferns as thin soil forms – wind-dispersed trees (birch) then animal-dispersed (oak, beech) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 73-76; AQA spec 3.1.3.3.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Environmental DNA (eDNA) analysis	1

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Transmitters are too heavy for small birds / their batteries would not last for a long-term study **1** Geolocator tags are small and only record time and light levels – latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each separate (regional) population has its own unique gene pool **1** DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Net held with its straight edge on the river bed facing the current (open to the flow) **1** The bed in front is disturbed by kicking / scraping a boot for a standard number of kicks and time (eg 30 s) **1** Same substrate type, water depth and flow rate at both sites – same net mesh – repeat samples – invertebrates identified with a key and counted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream	3
03.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Difficult to standardise the vigour / area kicked **1** Buried organisms or those fixed to stones are not dislodged **1** Some organisms swim and avoid the net **1** Flow rate varies and affects capture **1** Fragile organisms destroyed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** A fixed area of bed is enclosed by a frame so the sampled area is standardised – side nets reduce escapes **1** But only works in fast enough flow and is hard to set on rough substrates / deep water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream	2
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Pollutant / oxygen levels fluctuate (eg with flow, time of day) so one reading may miss a pollution event **1** Sensitive taxa (stonefly, mayfly larvae) disappear and tolerant ones (midge larvae, hoglouse) remain, integrating conditions over weeks / months (biotic index) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 395-396, 415-416; AQA spec 3.7.2.1.	
04.1	**1 mark each** 1 mark for each of the 4 points below. **1** Area must be large enough to support viable populations (breeding numbers high enough, gene pool large enough to avoid inbreeding) – species higher up food chains need larger areas (lower density) **1** Large areas contain a full range of resources (eg different fruiting trees / feeding areas through the year) **1 Shape:** the perimeter has edge conditions (a mix of neighbouring habitats) – a compact shape maximises core habitat for species needing core conditions **1** Some species benefit from edges (higher light) or small isolated habitats (ponds too small for predatory fish suit amphibians) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	4

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 3 points below. **1 Varied topography:** banks, lagoons, islands, ditches created with machinery give a range of water depths (emergent, floating-leaved, submerged plant zones) **1** Islands provide roosting / nesting sites safe from terrestrial predators (foxes) **1** Control of salinity (eg intertidal conditions for opossum shrimps feeding dunlin and avocet) – scrapes for skylarks – different water depths for waders and ducks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Mobile species (birds, flying insects) colonise easily **1** Less mobile species (many plants, amphibians, invertebrates) may not reach an isolated site so must be introduced – keystone species especially **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **keystone species** → a species whose loss changes the whole community	2
04.4	1 mark for the point below. **1** Allows dispersal of surplus young, mixing of gene pools (less inbreeding) and recolonisation if a population dies out – avoids hazards (eg rope bridges for howler monkeys over roads) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-42; AQA spec 3.1.2.3.4.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Overall decline from 62% to 22% **1** Sharp falls in 1998 (to 35%) and 2016 (to 18%) coinciding with bleaching events, with partial recovery after each **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 4 points below. **1** The polyps expel their symbiotic zooxanthellae so the coral loses colour – bleached corals are alive but grow very slowly and may die if algae do not return **1** Raised water temperature (climate change / El Niño) **1 Pollution:** sewage, pesticides, cyanide, sunscreen – low oxygen from zooplankton blooms after overfishing **1** High light / solar activity – ocean acidification hinders skeleton formation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiotic** → a close relationship in which both species benefit **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
05.3	**1 mark each** 1 mark for each of the 3 points below. **1** Coastal construction and dredging increase sedimentation / turbidity covering and killing polyps **1** Sewage and fertiliser nutrients cause algal growth that shades corals and prevents egg release **1** Physical damage from anchors, divers, souvenir collection – loss of mangroves / seagrass nurseries that trap sediment **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
05.4	1 mark for the point below. **1** Protected breeding populations grow larger and produce surplus young / adults that spill over into fished areas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 60-63; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** The new opportunity is overwhelmingly attractive (economic benefit) **1** Lack of knowledge / understanding of the impacts **1** The impacts occur elsewhere or much later, so are not evident to those who benefit **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Cultural belief that resources should be exploited to satisfy needs only **1** Problems of over-exploitation become obvious quickly as local supplies dwindle, so behaviour changes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Resources are brought in from other areas **1** If one area is depleted, exploitation moves to another area (eg timber, fish, metals, energy) without the consumers noticing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.4	**1 mark each** 1 mark for each of the 3 points below. **1** Easier access to energy has made many new activities possible **1** The scale of activities has greatly increased **1** Toxic wastes are produced that cannot be decontaminated by natural processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-371; AQA spec 3.6.1.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Sinkers / weighted lines so hooks sink quickly below the reach of albatrosses **1** Setting lines at night when birds are not feeding **1** Bird-scaring streamers / decoys **1** Curved hooks that catch tuna but not birds **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
07.2	**1 mark each** 1 mark for each of the 3 points below. **1 Effort:** limits on boat size, engine power, days at sea, number of licences – decommissioning payments to reduce fleet size where overfishing is unavoidable **1 Methods:** bans or length limits on drift nets – bans on demersal trawling over sensitive seabed (Rockall Bank, Hatton Bank deep-water reefs) **1** Shorter trawl tows so turtles / by-catch survive (over 99% survive under 10 minutes) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **decommissioning** → safely dismantling a power station at the end of its life **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	3
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Eggs / larvae are reared in captivity through the planktonic stage when most die (predation, failure to find habitat) **1** Juveniles are released when survival chances are good, increasing recruitment to the adult population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Egg / larval survival varies greatly between years so adult numbers do not predict recruitment **1** Several poor years would lower the MSY – egg surveys allow catches to be reduced in advance rather than after a sudden collapse **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **MSY** → the largest catch that can be taken without reducing the stock	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** Domesticated crops were bred from a small number of original plants so they have a small gene pool / low genetic diversity **1** Selective breeding for yield, appearance or shelf life may have lost other useful genes (eg taste, disease resistance) **1** Genetically similar plants share the same susceptibilities so a new disease or environmental change can affect the whole crop **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** A region with a high concentration of the wild relatives of important crops - usually where the crop was first domesticated (eg wheat / barley in the Middle East; potato, tomato in South America; rice in India; maize in Central America) **1** They hold the largest gene pools / most genetic variety for future crop breeding (disease resistance, drought and salt tolerance, yield) **1** Many are in regions where environmental degradation / habitat loss threatens the wild plants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	3
08.3	**1 mark each** 1 mark for each of the 3 points below. **1** A large population may all descend from a few closely related individuals so they are genetically similar **1** Similar adaptations mean all are susceptible to the same change (disease, climate) **1** Greater risk of inbreeding so disadvantageous recessive genes are expressed (eg amelanism, poor muscle development in lion cubs) – so species should be protected across their whole range to retain regional genes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3
08.4	1 mark for the point below. **1** Morama bean (drought resistant, soya-like protein) – yeheb tree (drought resistant nuts, poor soils) – potato bean (high protein) – perennial maize / Kernza (no annual replanting, less ploughing / erosion) – salt-tolerant spinach relatives **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 15-17; AQA spec 3.1.2.1.2.	
09.1	**Marks shared** Up to 6 marks in total - share them between the 5 points below according to how completely each one is made. **up to 2** Non-indigenous species (Sitka spruce from North America) chosen for fast growth and pest resistance – indigenous wildlife's needs for food, pollination, seed dispersal and habitat are not met **up to 2 Single species:** easy planting, thinning and felling over large areas – but each tree species supports its own community so few species overall **up to 2** Close planting makes trees compete for light so they grow tall and straight (good timber) but shades out ground flora and the species that depend on them **up to 2 Simple age structure:** uniform size makes management easy but gives little variety of light, temperature, wind – no old trees or dead wood for invertebrates and woodpeckers **up to 2 Mixed woodland:** varied niches, stratification, dead wood, indigenous trees each with their own dependent species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	6
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Trees are felled when income will be greatest - approaching full size when the growth rate starts to decline **1** Continuing to grow them ties up land and capital for little extra timber – natural forests keep old and dead trees **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Industrial society replaced species-specific uses (baskets, fencing, charcoal, tool handles) with large-scale uses: chipboard, plywood, pulp **1** Wood fuel replaced by fossil fuels / electricity – so traditional management declined and woods became uniform **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 365-367; AQA spec 3.5.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Why needed:** growing population (to 12 billion), land degradation since the fertile crescent, reliance on artificial fertilisers, pesticides and unsustainable irrigation – limits to new land and imports • **Pest control:** cultural methods (weeding, mulching, rotation, barrier / companion crops, predator habitats, biological control, sterile males, pheromones), resistant varieties, integrated control, reduced antibiotics • **Nutrients:** natural processes - nitrogen-fixing legumes, decomposition, recycling organic matter (manure, compost), crop rotation, permaculture, conservation of soil biota and mycorrhizae, reduced artificial fertiliser – phosphate recycling • **Energy:** natural processes instead of Haber-process fertiliser, low tillage, less machinery, renewable energy, low food miles, reduced methane (cattle diets, rice varieties) • **Water:** low water-use crops, drip irrigation, reservoirs and aquifer recharge, avoiding saline water – soil: organic matter maintenance, erosion control • **Biodiversity:** retaining hedgerows, ditches, ponds, woodlands – CWR conservation and seed banks – agri-environment schemes • **Social:** control of impacts on rural communities, Fairtrade, local food – consumer choices • **Evaluation:** yields vs sustainability, cost, need for policy support – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **sustainability** → meeting present needs without stopping future generations meeting theirs • **biological control** → using a natural predator, parasite or pathogen of the pest • **nitrogen-fixing** → nitrogen gas converted into ammonium or other nitrogen compounds **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	
	Sustainability matters because the population may reach 12 billion, land has been degraded since farming began in the fertile crescent, and current yields depend on artificial fertilisers, pesticides and irrigation that draws down groundwater, with little new land to cultivate. Pest control can move from chemicals to cultural methods - weeding, mulching, rotation, barrier and companion crops, habitats for predators - biological control, sterile males and pheromones, resistant varieties and integrated control, with fewer antibiotics in livestock.	
	Nutrients can come from natural processes: legumes fixing nitrogen, recycling manure and compost, rotation, permaculture and conserving soil biota and mycorrhizae, reducing manufactured nitrate and the non-renewable rock phosphate it accompanies. Energy use falls with low tillage, less machinery, renewable power, low food miles and lower methane from cattle diets and drought-tolerant rice. Water is conserved with low water-use crops, drip irrigation, reservoirs and aquifer recharge and by avoiding saline water; soil by maintaining organic matter and controlling erosion. Retaining hedgerows, ditches, ponds and woodland, conserving crop wild relatives in seed banks and agri-environment schemes protect the biodiversity that supplies pollination and pest control, while Fairtrade, local food and consumer choices address the social side.	
	The tension is between yield and sustainability: some strategies cost more or yield less in the short term and need policy support to be adopted. But since the unsustainable inputs themselves depend on finite fossil fuels and phosphate, the strategies that use natural processes are the only ones that can feed a larger population indefinitely.	
	Weaker answer - Level 2 (4 marks)	
	Agriculture can be made more sustainable by using less pesticides and using biological control instead, using manure instead of artificial fertilisers, growing legumes which add nitrogen to the soil, and using crop rotation. Drip irrigation saves water. Farmers can keep hedgerows to provide habitats for wildlife. Organic farming uses these methods. These strategies reduce pollution and keep the soil healthy so food can be produced in the future.	
	What makes the difference	
	• Level 2 lists valid strategies but does not explain why they are needed (population growth, land degradation, dependence on finite inputs) or group them by the problem they solve.	
	• Level 3 covers energy, soil, biodiversity and social strategies as well as pests, nutrients and water.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 discusses the yield-sustainability tension and the need for policy support; Level 2 has no evaluation.	
10.2	1 mark for the point below. **1** Agricultural systems designed to mimic natural ecosystems - perennial crops, mixed species, recycling of nutrients, minimal external inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Dynamic equilibria and negative feedback in climate, hydrological cycle and populations – positive feedback and tipping points caused by humans • **Energy:** solar-driven, low energy density, low temperature (photosynthesis, enzymes, nitrogen fixation vs Haber process) – renewable energy, heat pumps, bioleaching, fuel cells • **Materials:** cycles, few abundant elements, biodegradable non-toxic wastes vs fossil fuels and dispersed minerals – recycling, Cradle to Cradle, industrial symbiosis • **Diversity and stability:** rainforests / reefs vs agroecosystems – diverse energy mixes • Optimum vs maximum production – supporting ecosystems in agriculture, forestry, fisheries, urban design • Case studies and metrics – limits and judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-379; AQA spec 3.6.1-3.6.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Setting priorities:** IUCN Red List criteria, EDGE, endemic, keystone, flagship species, population dispersal • **Legislation and protocols:** Wildlife and Countryside Act, designated areas (SSSI to MCZ), CITES Appendices, IWC, EU CFP, ITTO - successes and limits of enforcement • **Captive breeding and release:** selection, difficulties, techniques, release site criteria, hard / soft release, monitoring – seed banks • **Habitat conservation:** creation, design (area, shape, corridors, age structure, colonisation), abiotic and biotic management, plagioclimax maintenance, population control, reintroductions • Habitat case studies (woodland, rainforest, coral reefs, islands, mangroves, Antarctica) illustrating threats and conservation efforts • Monitoring and new technologies informing decisions • **Evaluation:** examples of success (Iberian lynx, Hawaiian goose, Scilly rats, Yellowstone) and failure (ITTO, tigers) – cost, politics, climate change – judgement on the most effective combination	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-71; AQA spec 3.1.2.3.	
	Model answer - Level 5 (23 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Biodiversity is lost through direct exploitation, habitat destruction, introduced species, pollution and changes to abiotic factors, so its conservation needs methods that address each cause, at scales from a single species to international law. Their effectiveness varies with the threat, the species and the resources available.	
	Deciding what to conserve comes first. The IUCN Red List uses population size, area, decline rate and probability of extinction to prioritise species, and endemic, keystone, flagship and evolutionarily distinct species receive weight because their loss has wider consequences. This is effective at directing scarce resources, but it favours well-studied vertebrates over invertebrates and plants.	
	Legislation and agreements protect species and habitats. In the UK the Wildlife and Countryside Act, SSSIs, NNRs and Marine Conservation Zones, and in Europe SACs, SPAs and the Habitats Directive, restrict damaging activities; internationally CITES controls trade in threatened species, the IWC regulates whaling, the Ramsar Convention protects wetlands and the Rio Convention on Biological Diversity commits nations to conservation. Their effectiveness depends on enforcement: CITES has reduced trade in ivory and rhino horn but poaching continues where enforcement is weak; the IWC moratorium allowed some whale populations to recover but Japan and Norway continued whaling; designations protect against development but not against pollution or climate change crossing boundaries.	
	In-situ habitat conservation and management are the most fundamental methods because they keep species with the community and abiotic conditions they depend on. Management uses ecological understanding: maintaining plagioclimaxes such as chalk grassland and heathland by grazing or burning, coppicing and pollarding woodland, controlling water levels, light and grazing, designing reserves with the right area, shape, biological corridors and buffer zones, and creating new habitats such as wetlands in worked-out gravel pits. Case studies show it works - coppiced woodland restored dormice and butterflies, wetland creation increased bitterns - but reserves are often small and fragmented and need permanent funding.	
	Ex-situ methods rescue species for which in-situ protection has failed. Captive breeding faces problems of space, diet, breeding triggers, social groups and small gene pools, addressed with studbooks, artificial insemination, embryo transfer and cryopreservation; seed banks such as the Millennium Seed Bank store plant genetic diversity cheaply. The Arabian oryx, California condor and pool frog show captive breeding can bring species back, but only where protected habitat exists for hard or soft release with post-release monitoring, and it is expensive and works for few species.	
	Ecological monitoring - image databases, eDNA, tracking, satellites - measures whether any of this is working and has improved the targeting of effort. Overall, legislation is effective when enforced, habitat management is the most effective and widely applicable method, and captive breeding is a last resort that succeeds only in combination with habitat protection. The Living Planet Index continues to fall, however, which shows that conservation methods are winning individual battles while the drivers of loss - population, consumption, land-use change and climate change - remain unaddressed.	

Do not write outside the box

Question	Marking guidance	Marks
	What makes the difference	
	• The essay is structured by **type of method** and evaluates the effectiveness of each in turn, with named examples for both successes and failures.	
	• It draws on several specification areas (biodiversity, conservation legislation, habitat management, captive breeding, ecological processes, monitoring) - the synoptic breadth Level 5 requires.	
	• The conclusion ranks the methods and then steps back to the bigger picture (Living Planet Index, underlying drivers) - a supported judgement rather than a summary.	
	• A Level 3 essay would describe the same methods accurately but with fewer examples, little evaluation of enforcement or cost, and a conclusion that all methods are important.	

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